# Multi-Theorem Circle Configurations
## Answer Key

High-school Geometry

**Quick Answers**

1. 55
2. 96, 109
3. 12.6
4. 9
5. 125.7
6. 56
7. 5, 3, 5, 1, 9
8. 10
9. 55.9
10. 4, -3, 6, 37.7

**Curriculum**

**Level:** High-school Geometry
HSG-C.B.5 – *problems 1, 2, 3, 4, 5, 6, 8, 9*
HSG-GPE.A.1–A.3 – *problems 7, 10*
*Difficulty 2–5 of 5 across 18 tagged checks.*

**Common wrong answers**

6. *If they got 146: forgot that an angle inscribed in a semicircle is a right angle*

**1.** Circle $O$: central $\angle AOB = 110°$, $C$ on the major arc. Find $m\angle ACB$.

**Solution:** $\angle ACB$ is inscribed in the circle and $\angle AOB$ is the central angle standing on the *same* arc $AB$, so the inscribed-angle theorem applies: the inscribed angle is half the central angle. $m\angle ACB = \frac{1}{2}(110°)$. $\boxed{m\angle ACB = 55°}$

**2.** Cyclic quadrilateral $ABCD$ with $m\angle A = 84°$ and $m\angle B = 71°$. Find $m\angle C$ and $m\angle D$.

**Solution:** In a quadrilateral inscribed in a circle the opposite angles are supplementary, because each pair of opposite angles is inscribed on the two arcs that together make the whole circle ($\frac{1}{2}(360°) = 180°$). $\angle C$ is opposite $\angle A$, so $m\angle C = 180° - 84°$; $\angle D$ is opposite $\angle B$, so $m\angle D = 180° - 71°$. $\boxed{m\angle C = 96°}$

Then, for the second pair: $\boxed{m\angle D = 109°}$

**3.** Circle $O$, radius 9 cm, inscribed $\angle ABC = 40°$. Find the length of minor arc $AC$ to the nearest tenth of a centimetre.

**Solution:** Two theorems, in order. *Inscribed angle:* the central angle on arc $AC$ is twice the inscribed angle, so $m\angle AOC = 2(40°) = 80°$, and the minor arc $AC$ measures $80°$. *Arc* *length:* an arc of 80° is $\frac{80}{360}$ of the circumference,

$$\text{arc } AC = \frac{80}{360} \cdot 2\pi(9) = \frac{2}{9} \cdot 18\pi = 4\pi \approx 12.566$$

$\boxed{12.6 \text{ cm}}$

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**4.** Chords $AB$ and $CD$ meet at interior point $P$ with $AP = 8$, $PB = x$, $CP = 6$, $PD = 12$. Find $x$.

**Solution:** The intersecting-chords theorem says the two products of the pieces are equal: $AP \cdot PB = CP \cdot PD$, so

$$8x = 6 \cdot 12 = 72 \implies x = \frac{72}{8}$$

Check: $8(9) = 72$ and $6(12) = 72$ ✓ $\boxed{x = 9}$

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**5.** Flowerbed of radius 10 cm; sector $AOB$ has central angle 144°. Find the sector area to the nearest tenth of a square centimetre.

**Solution:** A sector of 144° is $\frac{144}{360} = \frac{2}{5}$ of the whole disc, so its area is that fraction of $\pi r^2$:

$$\text{area} = \frac{144}{360} \cdot \pi(10)^2 = \frac{2}{5}(100\pi) = 40\pi \approx 125.664$$

$\boxed{125.7 \text{ cm}^2}$

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**6.** $AB$ is a diameter of circle $O$, $C$ is on the circle, and $m\angle ABC = 34°$. Find $m\angle BAC$.

**Solution:** Two theorems again. *Angle in a semicircle:* $\angle ACB$ is inscribed in a semicircle (it stands on the diameter $AB$), so $m\angle ACB = 90°$. *Triangle sum:* $m\angle BAC = 180° - 90° - 34°$. $\boxed{m\angle BAC = 56°}$

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**7.** $x^2 + y^2 - 10x - 6y + 9 = 0$. (a) Centre-radius form, centre and radius. (b) The $x$-values where the circle crosses the $x$-axis.

**Solution (a):** Group and complete the square in each variable. Half of 10 is 5 and half of 6 is 3, so add $5^2$ and $3^2$ to both sides:

$$\begin{aligned}(x^2 - 10x) + (y^2 - 6y) &= -9 \\ (x^2 - 10x + 25) + (y^2 - 6y + 9) &= -9 + 25 + 9 \\ (x-5)^2 + (y-3)^2 &= 25\end{aligned}$$

The centre is $(h, k)$ read off with the opposite signs, and the radius is $\sqrt{25}$.

$\boxed{\text{centre } (5,3), \quad r = 5}$

**Solution (b):** A point on the $x$-axis has $y = 0$, so substitute $y = 0$ into the original equation:

$$x^2 - 10x + 9 = 0 \implies (x-1)(x-9) = 0$$

Both crossings are real because the distance from the centre to the $x$-axis is 3, which is less than the radius 5. $\boxed{x = 1 \text{ and } x = 9}$

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**8.** Tangent $PT = 12$ from exterior point $P$; a secant through $P$ meets the circle at $A$ then $B$, with $PA = 8$. Find $AB = x$.

**Solution:** The tangent-secant (power of a point) theorem says $PT^2 = PA \cdot PB$. The whole secant is $PB = PA + AB = 8 + x$, so

$$\begin{aligned} 12^2 &= 8(8+x) \\ 144 &= 64 + 8x \\ 8x &= 80 \end{aligned}$$

Check: $PB = 8 + 10 = 18$ and $8(18) = 144 = 12^2$ ✓ $\boxed{x = 10}$

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**9.** Circle $O$, radius 14 cm, $\angle AOB = 90°$. Find the area of the circular segment cut off by $AB$, to the nearest tenth of a square centimetre.

**Solution:** The segment is what is left of the sector once the triangle is removed, so this needs the sector formula *and* the triangle area. *Sector:* $\dfrac{90}{360}\pi(14)^2 = \dfrac{1}{4}(196\pi) = 49\pi \approx 153.938$ cm$^2$. *Triangle:* $\angle AOB$ is a right angle between two radii, so the triangle is right-angled with legs 14 and 14: area $= \frac{1}{2}(14)(14) = 98$ cm$^2$. *Segment:* $153.938 - 98 = 55.938$.

$\boxed{55.9 \text{ cm}^2}$

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**10.** Fountain $x^2 + y^2 - 8x + 6y - 11 = 0$ (metres). (a) Centre and radius. (b) Area of a 120° sector. (c) Explain the fraction.

**Solution (a):** Complete the square: half of 8 is 4, half of 6 is 3,

$$(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 \implies (x-4)^2 + (y+3)^2 = 36$$

so the radius is $\sqrt{36}$ and the centre is $(4, -3)$ — note the sign flip on the $y$ term: $(y+3)^2 = (y-(-3))^2$. $\boxed{\text{centre } (4,-3), \quad r = 6 \text{ m}}$

**Solution (b):** The watered region is a sector of the disc of radius 6:

$$\text{area} = \frac{120}{360} \cdot \pi(6)^2 = \frac{1}{3}(36\pi) = 12\pi \approx 37.699$$

$\boxed{37.7 \text{ m}^2}$

**Solution (c) — model answer (open response, review by hand):** A full turn about the centre is 360°, and the sprinkler sweeps 120° of it. Sectors with the same radius have areas in the same ratio as their central angles, because a sector of angle $\theta$ is exactly what you get by cutting the disc into $\frac{360}{\theta}$ congruent pieces and keeping one. Since $\frac{120}{360} = \frac{1}{3}$, the watered region is one third of the fountain's area — and that is why the formula multiplies $\pi r^2$ by $\frac{\theta}{360}$ rather than by $\theta$ itself.